

Ilia Malkin

Junior Python Engineer · LLM integrations & backend automation

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SUMMARY

Backend-focused Python engineer with a Data Science background (Universität Wien), building production LLM pipelines, agentic systems and multi-provider LLM routers with fallback and circuit breaker logic. No team experience yet; several open-source projects of varying depth back the resume instead. Strongest in LLM SDK integration (OpenAI, Gemini, xAI/Grok, MiMo, OpenRouter), async Python, prompt engineering, Pydantic-typed structured outputs and multi-provider fallback patterns. Looking for junior backend / AI-engineering roles. Remote only; on-site considered in Vienna / EU; no on-site work in Russia under any circumstances.

PROJECTS

Stager · multi-tenant expense tracker

May 2026 · github.com/IliaMalkin/stager

Telegram bot + FastAPI backend + Next.js admin for receipt-photo OCR via LLM

- Designed and shipped a 9-service Docker Compose stack (bot, api, worker, beat, postgres, redis, minio, web, caddy) on a single Hetzner VPS; full Alembic migrations, healthchecks, Caddy TLS.
- Built a custom LLM router (MiMo → Gemini fallback) with Redis-based soft circuit breaker, Pydantic-validated structured output, structlog metrics and per-request fallback-reason logging.
- Backend in FastAPI + Pydantic v2 + async SQLAlchemy 2.0 (PostgreSQL 16); JWT auth + RBAC (owner/editor/viewer); invite-token redeem flow.
- Bot in aiogram 3 with FSM stored in Redis: photo → OCR → review-card → inline-button edits → expense persistence.
- Async jobs via Celery + Redis (OCR, XLSX export with openpyxl); Celery Beat for scheduled tasks.
- Web admin in Next.js 14 (App Router) + Tailwind + recharts, JWT in httpOnly cookie, server actions for mutations.
- Test stack: pytest + testcontainers for Postgres/Redis paths + respx for LLM mocking; GitHub Actions runs ruff, mypy and tests; pre-commit + gitleaks locally.

doc-assistant-bot · Telegram RAG bot

2025 · github.com/IliaMalkin/doc-assistant-bot

Telegram bot that answers questions over uploaded documentation

- Single-process aiogram bot with SQLite + sqlite-vec vector search, fastembed embeddings and OpenRouter answer generation.
- Supports PDF, DOCX and TXT uploads, answers with source citations, has a seven-language UI, pytest coverage and GitHub Actions CI.

revenue-metrics-console

2025 · github.com/IliaMalkin/revenue-metrics-console

Full-stack SaaS metrics dashboard: React frontend + Node/Express API over PostgreSQL

- Full-stack TypeScript monorepo (pnpm workspaces): Node.js + Express + Drizzle ORM over PostgreSQL; 9 server-side SaaS metrics (MRR, ARR, LTV:CAC, NRR, churn), JWT auth (access + refresh) with RBAC, WebSocket live feed.
- React 18 frontend (TanStack Router/Query, Zustand, Chart.js): cohort retention heatmap, MRR waterfall, CSV export; tested with Vitest + Playwright, GitHub Actions CI.

b2b-outreach-pipeline

2026 · github.com/IliaMalkin/b2b-outreach-pipeline

AI-assisted outreach data workflow

- Python pipeline for company seed CSV → website-based personalization → masked CSV/XLSX export, email sequence and QA summary.

windows-claude-code-doctor

2026 · github.com/IliaMalkin/windows-claude-code-doctor

Claude Code skill for diagnosing AI agents on Windows

- PowerShell-based checks for broken Python environments, PATH conflicts, MCP-provider misconfiguration; published as a reusable Claude Code skill.

TECHNICAL SKILLS

Languages:	Python, TypeScript, JavaScript, SQL, PowerShell, Bash
Backend:	FastAPI, Pydantic v2, async SQLAlchemy 2.0, Alembic, aiogram 3, Celery, REST APIs, JWT, RBAC
LLM / AI:	OpenAI SDK, Google Gemini, xAI Grok (OAuth + Agent Tools), MiMo, OpenRouter, RAG (sqlite-vec, fastembed), prompt engineering, structured output, function calling, multi-provider routers with circuit breaker
Frontend:	React 18, Next.js 14 (App Router), Tailwind CSS, TanStack Query, recharts, Chart.js; Node.js/Express + Drizzle ORM (full-stack TS)
Data / Infra:	PostgreSQL 16, Redis, MinIO (S3), Docker, Docker Compose, Caddy, Hetzner, Linux, CSV/XLSX automation
Quality:	pytest, testcontainers, respx, ruff, mypy, pre-commit, gitleaks, structlog, Sentry
CI / Tools:	GitHub Actions, Git, Alembic, OpenAPI

EDUCATION

Universität Wien · BSc in Data Science	in progress
Statistics, machine learning, data processing, algorithms.	

LANGUAGES

Russian: Native · English: B2 · Deutsch: B2